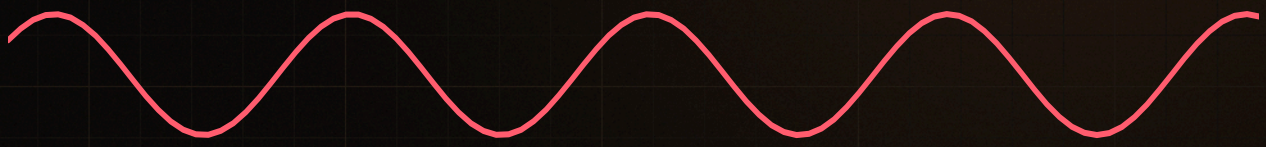
CHANNEL 06 · ACTIVE CH-06 · 10.5 GHz



GRID SYNCED // 10.48 GHz · 50.00 Hz · 416 VAC

CLASSIFIED — PROBLEM STATEMENT CHANNEL

ENERGY, POWER & SUSTAINABILITY

PROBLEM STATEMENTS · ROUND 1

24-HOUR NATIONAL-LEVEL HARDWARE HACKATHON

29–30 JANUARY 2027 · FR. C. RODRIGUES INSTITUTE OF TECHNOLOGY, VASHI

DOMAIN SLOT 06 CH-06 · 10.5 GHz



GRID SYNCED // 10.48 GHz · 50.00 Hz · 416 VAC

ENERGY, POWER & SUSTAINABILITY

The transition toward cleaner, resilient energy distribution, electrification and high-efficiency power management. Physical systems that optimize renewable capture, battery reliability and grid stability.

— AREAS TO EXPLORE

Smart Grid Telemetry | EV Charging & BMS | Solar / Wind MPPT | Energy Harvesting | Power Conditioning

— WHO SHOULD EXPLORE

Electrical Machines · Power Systems · Battery Storage · Renewable Energy & Embedded Control

> *"From raw generation to intelligent distribution — harness the current and power a sustainable tomorrow."*

— IN THIS CHANNEL

PS-01	Solar Panel Hotspot & Dust Accumulation Detection System
PS-02	Agricultural Pump Protection Against Dry Running & Voltage Fluctuations
PS-03	Distribution Transformer Overloading & Phase Imbalance Monitoring
PS-04	Smart Energy Management for Solar-Powered Buildings with Battery Backup
PS-05	Underground Power Cable Fault Detection & Distance Estimation

PS-01 SOLAR PANEL HOTSPOT & DUST ACCUMULATION DETECTION SYSTEM

Recommended for First & Second Year students

A · PROBLEM DEFINITION

Photovoltaic (PV) panels lose output when dust films their surface (soiling) or when individual cells overheat locally (hotspots) because of partial shading, cell cracks or mismatch. Both faults are hard to notice by eye and are usually discovered only through falling energy bills. The challenge is to detect and distinguish soiling and hotspot conditions early, using low-cost electrical and thermal measurements.

B · CONTEXT

Dust can cut the output of a panel by several percent within weeks in dry regions, while a persistent hotspot can permanently damage cells and, in extreme cases, cause fire. Today most owners rely on periodic visual checks or costly thermal-camera surveys, impractical for small rooftop and agricultural installations. A continuous, low-cost method that tells the owner when to clean and when to call a technician would raise energy yield, improve safety and reduce needless cleaning and water use.

C · CORE FEATURES

- Measure panel voltage, current and temperature, and compare against ambient conditions and a reference.
- Estimate soiling loss by comparing actual power with the power expected at the measured irradiance and temperature.
- Detect localised abnormal heating and differentiate it from uniform, panel-wide dust loss.
- Raise graded alerts (e.g. “clean soon” and “inspect urgently”) instead of a single alarm.
- Log data and show trends such as daily yield, performance ratio and temperature rise.

D · DESIGN REQUIREMENTS

- Users: rooftop owners, small solar farms and solar-pump installations; cost should be a small fraction of the panel price.
- Operation: withstand heat, humidity, dust and monsoon rain with minimal maintenance and very low self-consumption.
- Economics: trigger cleaning only when estimated loss crosses a justified threshold (e.g. 5–10%), saving labour, water and energy.
- Safety: early hotspot warning should reduce fire risk, and sensing must not compromise panel electrical safety.

E · TECHNICAL FOCUS

SENSING Panel voltage and current, back-sheet temperature (thermistor, RTD or digital sensor), ambient temperature and an irradiance reference (small reference cell, calibrated sensor or LDR-based estimate).

PROCESSING & LOGIC Any low-cost microcontroller. Irradiance-normalised power (performance ratio), temperature-difference thresholds and simple rule-based decisions are sufficient.

COMMUNICATION Local display or buzzer, plus an optional Wi-Fi, Bluetooth, LoRa or GSM link to a mobile or web dashboard; justify the choice for the site.

POWER SUPPLY Self-powered from the monitored panel or a small battery; state the power budget.

SIMULATION Model PV behaviour (e.g. single-diode model) in MATLAB/Simulink, Python or an equivalent tool; inject soiling, shading and hotspot cases and show detection.

OPTIONAL Thermal-image processing, drone survey, ML classification, automated cleaning trigger, string-level monitoring.

F · EXPECTED OUTCOME

- Block diagram, sensor placement plan and decision-logic flowchart.
- Simulation of three cases (clean, soiled, hotspot) showing measured signals and system response.
- Sample output plots: performance ratio against time, detection thresholds and alert generation.
- Discussion of false alarms, missed detections and the effect of weather variation.
- Bill of materials with estimated cost, and a note on scaling from one panel to a multi-panel array.

PS-02 AGRICULTURAL PUMP PROTECTION AGAINST DRY RUNNING & VOLTAGE FLUCTUATIONS

Recommended for First & Second Year students

— A · PROBLEM DEFINITION

Agricultural pump sets (typically 1–10 HP induction motors) run unattended in remote fields on a weak, fluctuating rural supply. If the water level drops and the pump runs dry, seals and windings overheat; low or high voltage, phase loss and phase unbalance overheat the motor in the same way. The challenge is to protect the pump automatically and restart it safely once conditions return to normal.

— B · CONTEXT

Pump motor burnout is a frequent and expensive problem for farmers, causing repair costs, lost irrigation time and crop stress. Conventional starters rely on thermal overload relays that react only after heating has begun and cannot recognise a dry well. Farmers are often far from the pump and cannot respond quickly. An affordable protection unit that senses abnormal conditions early, trips the motor within seconds and restarts it intelligently would prevent damage and make irrigation more reliable.

— C · CORE FEATURES

- Sense supply voltage on all phases and detect undervoltage, overvoltage, phase loss, phase reversal and unbalance.
- Detect dry running through a motor current or power-factor signature and/or a water level or flow sensor.
- Trip the contactor within a defined time and show the fault type locally (LED, buzzer or display).
- Restart automatically after a delay with a retry limit, avoiding rapid on-off cycling.
- Optionally notify the farmer of trips and restarts remotely.

— D · DESIGN REQUIREMENTS

- Affordability: cost a small fraction of a motor replacement, simple enough for a non-technical farmer to install and reset.
- Robustness: tolerates dust, moisture, voltage surges, heat and insects, with minimal maintenance.
- Connectivity: must work fully without internet access; remote alerts are a bonus, not a necessity.
- Safety & standards: isolated sensing, correctly rated contactor and compliance with applicable IS/IEC electrical safety norms.

— E · TECHNICAL FOCUS

SENSING Voltage sensing (potential transformer or isolated divider), motor current (CT or Hall sensor) and, optionally, a float, electrode or pressure-based level sensor.

PROCESSING & CONTROL Microcontroller or discrete logic driving a relay/contactor, with timers, hysteresis and fault latching. Justify trip thresholds (e.g. ±10–15% of nominal voltage) and trip delays.

COMMUNICATION Optional GSM/SMS, LoRa or Bluetooth; local fault indication is mandatory.

POWER SUPPLY Isolated supply that tolerates wide voltage swings; consider behaviour during phase loss.

SIMULATION Motor and load model in MATLAB/Simulink, Proteus, LTspice or Python covering healthy, dry-run, low-voltage and single-phasing scenarios; report trip times.

OPTIONAL Soft starter, solar-pump compatibility, level-based automatic start/stop, run-hour and energy logging.

— F · EXPECTED OUTCOME

- Block diagram and protection-logic flowchart with justified thresholds.
- Simulation of at least four scenarios (normal, dry run, over/undervoltage, phase loss) with measured trip times.
- Demonstration that the restart logic prevents repeated start-stop cycles.
- Estimated cost and a comparison with a conventional starter or overload relay.
- Notes on installation, field limitations and possible improvements.

PS-03 DISTRIBUTION TRANSFORMER OVERLOADING & PHASE IMBALANCE MONITORING

Recommended for Second & Third Year students

A · PROBLEM DEFINITION

A distribution transformer (typically 11 kV/415 V, 100–500 kVA) supplies local consumers. When single-phase loads are spread unevenly across the three phases, one phase is overloaded, neutral current rises and the windings heat unevenly; sustained overloading also ages the insulation. The challenge is to monitor loading and phase imbalance continuously and warn before damage or an outage occurs.

B · CONTEXT

Utilities manage thousands of transformers, yet most have no real-time visibility on the low-voltage side. Overloading and imbalance raise losses, shorten insulation life and cause failures that cut supply to entire neighbourhoods. Periodic manual readings miss evening peaks and seasonal surges. A low-cost monitoring node that measures loading, imbalance and thermal stress would enable timely load balancing, capacity upgrades and fewer failures.

C · CORE FEATURES

- Measure three-phase currents and voltages on the LV side and compute loading as a percentage of rated kVA.
- Calculate phase unbalance and neutral current.
- Track tank/oil and ambient temperature to estimate thermal stress and approximate loss of insulation life.
- Issue severity-based alerts (warning, critical) for sustained overload, high imbalance or overtemperature.
- Record load profiles and generate daily and monthly reports to support load-balancing decisions.

D · DESIGN REQUIREMENTS

- Users: electricity distribution companies, industrial campuses and housing societies.
- Cost & rollout: low cost per node and non-intrusive installation (clamp-on sensing, no shutdown) to be viable across thousands of transformers.
- Environment: outdoor-rated, tolerant of heat, vibration, dust and surges, workable in poor mobile coverage.
- Value & safety: data should support load balancing, capacity planning and loss reduction; all sensing is on the LV side only.

E · TECHNICAL FOCUS

SENSING Split-core current transformers on each phase and the neutral, isolated voltage sensing and a temperature sensor on the tank or oil pocket.

PROCESSING & ANALYTICS Microcontroller or energy-metering IC computing RMS values, kVA, power factor and unbalance (percentage deviation from average, or sequence components). A simplified thermal model (IEC 60076-7 or IEEE C57.91) may be used for loss-of-life estimates.

COMMUNICATION GSM, NB-IoT, LoRa or similar, with local buffering (store-and-forward) for patchy coverage.

POWER SUPPLY Powered from the LV supply with a small backup so outage events are still reported.

SIMULATION Three-phase unbalanced load model with 24-hour load profiles and a thermal model in MATLAB/Simulink, Python or equivalent; demonstrate alert generation.

OPTIONAL Load forecasting, harmonic analysis, oil-level sensing, tamper or theft indication, tap-position monitoring.

F · EXPECTED OUTCOME

- System architecture showing sensor placement and data flow.
- Simulation of balanced, unbalanced and overloaded cases with computed loading, unbalance and temperature estimates.
- Alert logic with justified thresholds and example alert outputs.
- Analysis of a sample 24-hour load profile ending in a load-balancing recommendation.
- Cost per node and a rollout/scalability plan for a distribution network.

PS-04 SMART ENERGY MANAGEMENT FOR SOLAR-POWERED BUILDINGS WITH BATTERY BACKUP

Recommended for Second & Third Year students

A · PROBLEM DEFINITION

A building with rooftop solar, battery storage and a grid connection must continuously decide which loads run on solar, when to charge or discharge the battery and when to draw from the grid. Poor decisions raise bills, wear out the battery or leave too little reserve during an outage. The challenge is to design an energy-management strategy that balances cost, solar self-consumption, battery life and backup reserve.

B · CONTEXT

Rooftop solar with batteries is spreading across homes, schools and small commercial buildings, but most installations run on fixed rules: charge whenever the sun shines and discharge whenever the load appears. These rules ignore time-of-day tariffs, battery stress and the importance of different loads. A smarter controller can shift flexible loads, protect the battery and guarantee reserve energy for critical loads, improving savings and reliability without extra generation capacity.

C · CORE FEATURES

- Monitor PV generation, building load, battery state of charge (SoC) and grid availability.
- Classify loads as critical, flexible or deferrable and manage them by priority.
- Schedule battery charging and discharging within safe SoC, current and temperature limits.
- Hold a configurable backup reserve and switch to island operation for critical loads during outages.
- Display energy flows, savings and battery status on a simple dashboard.

D · DESIGN REQUIREMENTS

- Users: homes, schools, clinics and small offices; payback should be competitive with simple controllers.
- Tariffs: support time-of-day pricing, net metering and sanctioned-load or peak-demand limits.
- Battery life: limit depth of discharge and charge rates, since replacement cost dominates lifetime economics.
- Safety & usability: battery protection, anti-islanding awareness for grid-tied systems and a simple user interface.

E · TECHNICAL FOCUS

SENSING AC/DC voltage and current at PV, battery, grid and key load branches; battery temperature; SoC estimation by coulomb counting and/or open-circuit voltage.

CONTROL STRATEGY Rule-based or optimisation-based scheduling (e.g. cost minimisation over a 24-hour horizon); justify the approach and its computational demand.

COMMUNICATION Modbus, MQTT, Wi-Fi or similar between the controller, inverter, meters and dashboard.

DATA & MODELS Daily solar irradiance, load and tariff profiles (synthetic or published datasets); inverter and charger may be simplified blocks.

SIMULATION Energy-flow simulation in MATLAB/Simulink, Python or equivalent over 24 hours to 7 days, compared against a simple baseline controller.

OPTIONAL Solar and load forecasting, EV charging coordination, live tariff data, demand-response participation.

F · EXPECTED OUTCOME

- System architecture and a flowchart of the energy-management algorithm.
- Simulated 24-hour and multi-day results showing SoC, grid import/export and load coverage.
- Quantified comparison with a baseline: cost savings (%), self-consumption (%) and battery cycling.
- A grid-outage scenario showing that critical loads are served for the claimed duration.
- Estimated implementation cost and payback analysis for a representative building.

PS-05 UNDERGROUND POWER CABLE FAULT DETECTION & DISTANCE ESTIMATION

Recommended for Third & Final Year students

A · PROBLEM DEFINITION

Underground power cables (low- and medium-voltage, for example 11 kV XLPE) are hidden from view, so faults such as short circuits, earth faults, open conductors and insulation breakdown cannot be located by inspection. The challenge is to detect and classify the fault and estimate its distance from a cable end using electrical measurements, despite fault resistance, noise, joints and non-uniform cable sections.

B · CONTEXT

Locating a buried cable fault by trial excavation or repeated re-energising is slow, costly and risky, and it prolongs outages in cities and industrial campuses. Established methods (impedance-based calculation, travelling-wave reflection timing and bridge methods) each have limits: fault resistance distorts impedance estimates, travelling waves need fast sampling, and specialised equipment is expensive. A practical, accurate and affordable approach would greatly shorten restoration time.

C · CORE FEATURES

- Detect a fault and classify its type (line-to-ground, line-to-line, double line-to-ground, three-phase, open circuit).
- Estimate the distance to the fault from measurements at one cable end, or at both ends if justified.
- Remain robust to fault resistance, measurement noise, load current and cable-parameter uncertainty.
- Report an estimated location together with an uncertainty or confidence indication.
- Capture and store event waveforms and generate a fault report for the maintenance crew.

D · DESIGN REQUIREMENTS

- Users: utilities, municipal bodies, metro and rail systems and industrial campuses.
- Value: shorter outage restoration and fewer excavation attempts; propose and justify an accuracy target (e.g. within a few percent of cable length).
- Affordability: substantially cheaper than commercial portable fault-location sets, installable at substations or carried by field crews.
- Safety: measurement through CTs/PTs or isolated, rated interfaces; no direct high-voltage contact by the operator.

E · TECHNICAL FOCUS

SENSING Current and voltage via CT/PT (sending end, optionally receiving end). For travelling-wave methods, sampling must resolve the distance: distance = $v \times \Delta t / 2$, with propagation velocity v of roughly 150–170 m/µs in typical XLPE cable.

PROCESSING Microcontroller, DSP or FPGA. Candidate algorithms include fundamental-frequency impedance estimation, wavelet or correlation-based reflection timing, and bridge-type (Murray loop) calculation; combinations are allowed.

COMMUNICATION & TIMING Event reporting over GSM/fibre/LoRa; if both-end methods are used, explain time synchronisation (e.g. GPS).

CABLE & FAULT MODELS Lumped (π-section) or distributed-parameter cable models with adjustable length, fault location, fault type and fault resistance.

SIMULATION MATLAB/Simulink, PSCAD, Python or equivalent. Include noise and at least one joint or section change.

OPTIONAL ML classifiers, both-end synchronised measurement, partial-discharge or cable-ageing indicators, GIS-based mapping of the estimated location.

F · EXPECTED OUTCOME

- System architecture, signal-processing chain and a clear description of the chosen algorithm.
- Simulated fault study covering at least three fault types and five or more fault locations.
- Error statistics (mean and maximum distance error, as a percentage of cable length).
- Sensitivity analysis to fault resistance, noise level and cable-parameter error.
- Hardware architecture, estimated cost and limitations compared with commercial fault locators.